

The latter part of the 20th century saw a return to the fascination with “flying like a bird” that had inspired the early flight pioneers. The growth of hang-gliding as a leisure activity allowed thousands of people to, in effect, repeat the experiences of Otto Lilienthal, while the microlight, essentially a hang-glider with an engine attached, is a remote descendant of the powered glider Percy Pilcher was working on when he died in 1899. In the 1970s, it even proved possible to fulfill the dream of human-powered flight, long dismissed by most as a physical impossibility.

Human-powered flight

As early as 1680, Italian mathematician Giovanni Borelli had calculated that human muscles could not generate enough power for birdlike flapping flight. “It is impossible,” he wrote, “that men should be able to fly craftily, by their own strength.” Until very recently, Borelli seemed to be right. Several cash prizes were put up in the 1930s for the first controlled, human-powered flight, but none was claimed. A human athlete is capable of generating around 0.4hp for sustained periods; the Wright brothers’ first aero-engine had generated 12hp. Manpower, it seemed, was just not power enough.

In 1959 a British industrialist, Henry Kremer, set out to reignite interest in human-powered aircraft with a substantial cash prize for anyone who could fly a figure-eight course around two pylons at least 2,625ft (800m) apart. Regularly increased to keep pace with inflation, the prize stimulated numerous experiments, but remained unclaimed for 18 years. However, in 1976 a team led by aeronautical engineer Dr. Paul MacCready of Pasadena,

IMITATING DAEDALUS

ACCORDING TO LEGEND, the first human-powered flights were made by Daedalus and his son Icarus, escaping from the island of Crete. In 1984, professors and students at the Massachusetts Institute of Technology (MIT) decided to see if they could build an aircraft to match this legendary feat. They assumed that the goal of the flight had been the smaller island of Santorini, about 74 miles (118km) north of Crete. Their design followed a pattern by then well established – lightweight materials, a long narrow wing, and a propeller driven by pedalpower. It still took four years and millions of dollars to create an airplane that might manage the distance. *Daedalus* took off from Crete’s Heraklion airport early on the morning of April 23, 1988, with Greek champion cyclist Kanellos Kanellopoulos in the

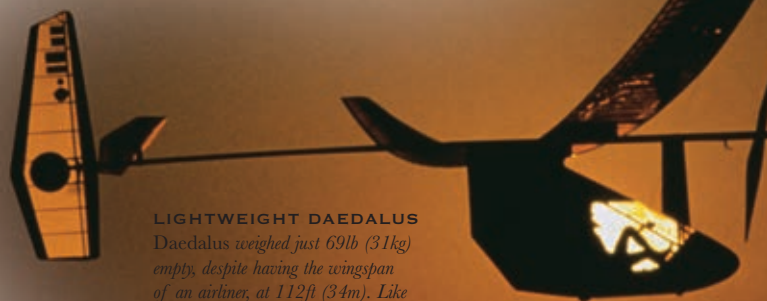


FLYING WING

Daedalus, the human-powered aircraft that made the record-breaking flight from Crete to Santorini, was created by staff and students at MIT.

cockpit. Fortunately aided by a following wind, he headed across the sea at about 20mph (32kph), flying some 16ft (5m) above the waves. The journey to Santorini took 3 hours 54 minutes and

proceeded without mishap until the final approach to land on a beach, when a sudden gust of headwind broke up the aircraft a few yards off shore. A gleeful Kanellopoulos emerged unscathed to celebrate a record for distance and endurance.



LIGHTWEIGHT DAEDALUS

Daedalus weighed just 69lb (31kg) empty, despite having the wingspan of an airliner, at 112ft (34m). Like the Gossamer series, it used pedal-power to drive a propeller.

California, decided to take up the challenge. Starting from scratch, with no reference to current powered-aircraft designs, they built an aircraft called the *Gossamer Condor*. It was made of light, plastic-covered aluminum tubes and its long, narrow wing was braced with stainless-steel wires, the leading edge reinforced with corrugated cardboard and styrene foam. Intriguingly, it shared some essential features with the Wright Flyer – a canard stabilizer at the front, a pusher propeller, and wing warping for lateral control. The pilot powered the aircraft by pedaling.

By the summer of 1977, MacCready’s team was ready for an attempt at winning the Kremer prize, which by then stood at £50,000 (\$90,000). The pilot was to be Bryan Allen, a champion cyclist and hanggliding enthusiast. In Shafter, California, on August 23, 1977, Allen powered

the *Gossamer Condor* along a figure eight course for a distance of 1½ miles (2.15km) at a speed of around 11mph (17.5kph). The MacCready team was able to carry off the prize because the aircraft was so light – the pilot weighed more than the machine – and so perfectly designed to maximize lift and minimize drag. When Kremer put up an even larger prize for the first human-powered flight across the Channel – shades of Blériot – the MacCready team won that too. With Allen again as pilot, on June 12, 1979, the *Gossamer Albatross* flew from Folkestone, England, to Cap Gris-Nez in France, a distance of 22 miles (35km).

MacCready subsequently turned his attention to improving the performance of his human-powered aircraft with a little help from the Sun. By placing solar cells on the surface of the wing, he was able to generate electricity to assist the pedaling pilot in his grueling task. In July 1981, driven by a blend of electric power and the leg muscles of pilot Steve Ptacek, MacCready’s